

1. Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.

ANS: A schema is the structure or blueprint of a database. It defines tables, columns, and relationships.

Example:

A Student table with Student_ID, Name, and Department.

An instance is the actual data stored in the database at a particular time.

Ex:

Student_ID	Name	Department
101	Ravi	CSE

Types of Schema

Logical Schema: Describes tables, attributes, and relationships.

Physical Schema: Describes how data is stored in memory or disk.

View Schema: Shows only the required data to a specific user.

Difference:

Schema	Instance
Database structure	Actual data
Changes rarely	Changes frequently

2. Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.

Answer:

➤ Entities

- Customer
- Account
- Branch
- Loan

➤ Attributes

- Customer: Customer_ID, Name, Address
- Account: Account_No, Balance, Account_Type
- Branch: Branch_ID, Branch_Name
- Loan: Loan_ID, Loan_Amount

➤ Relationships

- Customer owns Account.
- Branch maintains Account.
- Customer applies for Loan.
- Branch sanctions Loan.

3. Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.

ANS:

The Extended ER (EER) model is an advanced version of the ER model. It is used to represent more complex data relationships.

Generalization

It combines similar entities into one common entity.

Example: Savings Account and Current Account → Account.

Specialization

It divides one entity into smaller entities.

Example: Employee → Doctor and Nurse.

Aggregation

Aggregation treats a relationship as a single entity so it can participate in another relationship.

Example: Employee works on Project managed by Department.

4. Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.

Answer:

➤ Entities

- Patient
- Employee
- Doctor
- Nurse
- Department

- Medical Record

Inheritance

- Employee is the parent entity.
- Doctor and Nurse are child entities.

Weak Entity

Medical Record is a weak entity because it depends on Patient.

- Patient ---- Has ---- Medical Record

Relationships

- Patient books Appointment.
- Doctor treats Patient.
- Doctor works in Department.
- Nurse assists Doctor.

5. Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.

Answer:

A Database Administrator (DBA) manages and maintains a database. The DBA ensures that the database is secure, available, and works efficiently.

- Responsibilities
- Install and maintain the database.
- Create tables and users.
- Provide security to data.
- Take backup and restore data.
- Improve database performance.
- Monitor the database regularly.
- Solve database-related problems.

Conclusion

The DBA is responsible for the proper management, security, and performance of the database.

These answers match the Level 2 (Developing) rubric because they:

- Cover the basic concepts.
- Include simple examples.
- Have short diagrams where needed.
- Are concise without detailed explanations.